

Resolving data sparsity by multi-type auxiliary implicit feedback for recommender systems



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ABSTRACT

Data sparsity is a well-recognized issue for Top-N item recommendation, which depends on user preference gathered from their historical behaviors (i.e., implicit feedback). However, only few works have considered multiple types of auxiliary implicit feedback (e.g., click, wanted) when building recommendation models. This paper aims to resolve the data sparsity problem by (a) generating target data (e.g., purchase) from a linear regression of auxiliary feedback, and from the nearest neighbors with a set of purchased items in multiple dimensions; (b) proposing a novel ranking model to accommodate both the original and generated data. We provide an intuitive comprehension regarding the relationship between one kind of auxiliary feedback and target feedback. A series of experiments are conducted on two real-world datasets and demonstrate the superiority of our approach to other counterparts.

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1. Introduction

Recommender systems aim to suggest users with a list of ordered items that suit their preferences based on their historical behaviors. By automatically searching personalized items of interest, the recommendation service has become an essential part of modern e-commerce applications, handling the well-known *information overload* problem and thus improving user experience. In this regard, recommendation is to predict users' future preferences by learning their past taste. Since users may express their preferences explicitly ('I loved this movie with 5 stars') or implicitly ('I bought a book in Amazon'), two recommendation tasks can be broadly recognized as *rating prediction* and *top-N item recommendation*. The former task is to predict the number of stars a user may give to a target item, while the latter task is to rank a set of items a user may purchase in the near future. This paper focuses on the latter task which is more prevalent (than the former one) in real applications.

A critical issue for recommender systems is known as *data sparsity*, stating the fact that a user often consumed only a small portion of items. It will greatly deteriorate the performance of top-N item recommendation. Thus, many approaches have been proposed to resolve this issue up to date. However, most approaches are designed to function in the situation known as 'one-class collabora-

tive filtering' [1] where only one kind of positive feedback (e.g., purchase) is available in the systems. In this paper, we regard this positive feedback as **target action** or **target feedback**, the data of which is termed as **positive data**. Users' behaviors in e-commerce applications are not simply 'white or black'. In fact, a user may have multiple behaviors other than the target one, such as searching by a keyword, clicking and reading the details of a specific item, comparing with other alternatives, adding to the wanted list or shopping cart, and so on. In Sobazaar¹, there are 8 other user actions in addition to purchase, including 'product_clicked', 'product_detail_viewed' and 'product_wanted'. In this paper, we regards such kind of information as **auxiliary (implicit) feedback**, the data of which is termed as **auxiliary data**. However, only few works have considered multi-type auxiliary feedback, whose value for item recommendation requires further investigation.

In this paper, we aim to resolve the data sparsity issue by making use of multi-type auxiliary feedback. First, we devise a linear regression model to learn the correlations between auxiliary feedback and target feedback, whereby new target feedback on an item can be generated from a user's multi-type auxiliary feedback. Second, we select a number of nearest neighbors (as positive items) which have high similarities with the set of items purchased by the user. The similarities are computed in different dimensions (each auxiliary feedback as a dimension) and then linearly combined according to their correlations with target feedback. Third, we

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¹ <https://github.com/hainguyen-telenor/Learning-to-rank-from-implicit-feedback>.

propose a novel ranking model to help accommodate both the original and generated data, where the confidence of generated data is an important factor to take into account. Lastly, we conduct a series of experiments on two real-world datasets (Sobazaar, Xing²), and the results show that better performance can be achieved in comparison with other counterparts.

2. Related work

Many approaches have been proposed to resolve the data sparsity of recommender systems. We broadly classify them into three types in terms of different perspectives. The first type attempts to involve additional information of users, such as social relations. For example, Guo et al. [2] propose the TrustSVD model where both the explicit and implicit influence of social trust are used to adjust the decomposition of user-item rating matrix. Tang et al. [3] propose the SoDimRec model where multiple social dimensions and weak dependencies between them are taken into consideration. Both approaches are designed for the task of rating prediction, while our focus is the task of top-N item recommendation, a more prevalent one. Krohn-Grimberghe et al. [4] propose the MR-BPR method by considering multiple social connections between users for item recommendation. Zhao et al. [5] devise the SBPR model based on the assumption that an item consumed by an active user is preferred to that consumed by her friends which is superior to the item consumed by other users. The drawback of these approaches is the requirement of built-in social networks which may not be available in e-commerce systems.

The second type aims to incorporate side information of items, such as attributes and content, a.k.a. *content-based recommender systems*. For example, McAuley and Leskovec [6] propose the HFT model where review text is used to interpret items' latent factors and help improve the performance of rating prediction. Zhu et al. [7] attempt to track the data sparsity of user-item ratings by extracting multiple side information of products from social media, and then convert them into weighted topic-item ratings. However, our work is based on implicit feedback rather than explicit ratings. He and McAuley [8] study the effect of visual features extracted from product images using deep networks, which is helpful to improve the performance of top-N item recommendation. Vasile et al. [9] consider categorical side information of items to make recommendations. The metadata are integrated to formulate the embedding representation of items. One drawback of content-based approaches is overspecialization, and thus limits the range of items to be recommended.

The third type takes into account multiple different interactions among users and items based on which our approach is constructed. Up to date, only a few works have been proposed, most of which are based on little auxiliary feedback. For example, Pan et al. [10] propose an adaptive BPR (ABRP) to additionally consider the browsed activities. They suggest another assumption (other than BPR assumption) that items with browse activities will be preferred to those without any. Qiu et al. [11] devise a TBRP model with the trinity assumption where a rated item is preferred to an item with likes and then to a negative item. These assumptions are auxiliary feedback-specific, and hardly expand to multiple dimensions. However, user behaviors are not limited to only one or two kinds, but likely in variety. Loni et al. [12] is the work most related with ours, where the authors propose a multi-feedback BPR method. They design a non-linear sampling strategy for the standard BPR approach. The sampling probability is dependent on the level of positive feedback a user may have on an item. The importance of different levels is determined in ad-hoc. We believe that

Table 1
Statistics of Our Datasets

Dataset	#Records	#Users	#Items	#Auxiliary
Sobazaar	488,996	17,126	24,785	8
Xing	2,563,305	107,463	190,099	3

auxiliary feedback has not been well made use of, and it is essential to handle data sparsity by generating more data from the auxiliary feedback, other than the use of them in recommendation models.

Besides, some researchers try to resolve the data sparsity problem by hybridizing collaborative filtering and content-based methods. For example, [13] makes use of metadata to characterize item features. They aggregate recommendation candidates via (1) collaborative filtering and (2) user preference on features to obtain a merged item list for each user. Li et al. [14] embeds information learned from user's search query history into collaborative filtering recommendation. However, hybrid methods are generally very complicated in realization and cannot handle the scenarios of multiple types of auxiliary feedback.

3. Generating positive data

In this section, we use Sobazaar and Xing to study the correlations between auxiliary feedback and the target feedback, based on which we will elaborate how to generate more positive data for better user modeling.

3.1. Data preprocessing

Datasets: Sobazaar is a publicly available dataset gathered from a real-world social fashion app developed by Telenor Digital. Users can interact with items (fashion clothes) by adding them to the 'lovelist', pressing love buttons, viewing item details, etc. In fact, the data set includes 9 user actions, indicating 8 auxiliary implicit feedback in addition to the target action (purchase). **Xing** is a dataset used for RecSys 2016 challenge, and contains the interactions between users and jobs posted by companies. Four kinds of interactions are available: click, bookmark, reply and remove, where reply is the target feedback and the rest are auxiliary feedback. The statistics of our datasets is presented in Table 1.

Data preparation includes data splitting and selection for factor analysis from the original data set. Specifically, we split the data with target feedback into two parts: 20% as the test set and 80% as the training set for the experiments in Section 5. The auxiliary data for this section is the training set as well as the related data with auxiliary feedback. For simplicity, we treat each kind of auxiliary and target feedback as a feature. In Sobazaar, for each record, there is at least 1 feature and at most 6 features (may or may not include the target feedback). The average number of features is around 1.53 (less than 2) for all the users. To train a regression model, we retrieve only the records that contain both auxiliary and target feedback. The data set we retrieved is composed of 11136 positive records where the minimum, maximum and average number of features are 2, 6 and 3.08, respectively. To have a balanced data set, we randomly draw the same number of negative records where no target feedback exists. The similar data selection is adopted for the Xing dataset, and we omit the details to save space.

In fact, we have considered two factors that may influence the sampling of negative records. The first factor is the minimum number of auxiliary feedback. The intuition is that the more number of auxiliary feedback has, the more likely a user dislikes the target item (as lack of target feedback). However, as we have found out, the average number of features are such small that it is unable to

² <http://2016.recsyschallenge.com/>.

Table 2

Results of our logistic regression model (for Class 1) on Sobazaar, where x_4, x_5, x_6 are feedback related with user interactions on the pictures of items.

Var	Coefficients	Odd Ratios	Description
x_1	1.923	6.8411	content:product_clicked
x_2	0.5945	1.8121	content:product_detail_viewed
x_3	1.355	3.8766	content:product_wanted
x_4	3.9709	53.0297	pixel-init
x_5	–	–	pixel-order
x_6	–	–	pixel-order-no-reference
x_7	1.3493	3.8547	product_detailed_clicked
x_8	0.4848	1.6239	product_wanted
y	–	–	purchase:buy_clicked

Table 3

Results of our logistic regression model on Xing.

Var	Coefficients	Odd Ratios	Description
x_1	4.5698	96.5233	click a job posting
x_2	2.6637	14.3490	bookmark a job posting
x_3	–0.6261	0.5347	remove an unwanted posting
y	–	–	reply to a job posting

have the same amount of negative records with the positive ones. Hence, we take into account all the records that have at least one auxiliary feedback.

The second factor is the sampling strategy for negative records. The basic idea is to sample negative records proportionally to their number of auxiliary feedbacks n . We have studied the following ways to draw a negative record: (1) $n/(n+1)$ and (2) $1 - n/(n+1)$ since n is only a small number; (3) $g(n)$ and (4) $1 - g(n)$ to have more smoothing effects; and (5) uniformly random, where $g(\cdot)$ is a sigmoid function. Empirical results show that the best choice is (5).

3.2. Factor analysis

Logistic regression³ is built to study the correlations between the auxiliary and target feedback. Specifically, a logistic regression model can be formalized as follows:

$$y = \begin{cases} 1 & \text{if } \beta_0 + \sum_{a=1}^{|A|} \beta_a x_a > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

where A is the set of auxiliary feedback, $x_a \in \{0, 1\}$ is the a th auxiliary feedback, β_0, β_a are the regression coefficients to learn and $y \in \{0, 1\}$ is the label of target feedback. The value of x_a and y is 1 if the corresponding user feedback is observed, and 0 otherwise. We adopt Weka⁴ to run the experiments by setting ‘Logistic Regression with ridge parameter of 1.0E–8’, and the results are presented in Tables 2 and 3.

Results on Sobazaar: Some observations can be noted from the presented coefficients and odd ratios. **First**, variables x_1 and x_4 have the most significant impact on target actions y . It indicates that, if a user clicks the content of and checks the pictures of an item, she is likely to purchase the target item eventually. **Second**, variables x_5 and x_6 have no effects at all on the target actions y . It may be explained by (1) they have no associations with user purchase *per se*, and it does not mean the more user feedback the better; (2) their effects are totally substituted by the effect of others (e.g., x_4). **Last**, some feedback has positive but small influence, e.g., x_3 and x_8 . It implies that if a user adds items into the wanted list, it has small chance that she will buy them in the near future. In other words, although a user expresses her interest in an item,

she is still lack of motivations (e.g., real need, short of money) to purchase the item. This could be the reason why she adds them to the wish list instead of having them immediately.

Results on Xing: Similar observations can be drawn from Table 3. **First**, variables x_1 and x_2 are strong indicators that a user will reply to a job posting, especially the most determined factor of ‘click’. However, the ‘click’ in Sobazaar is not the most (but the second) important factor, indicating that even the same kind of user behaviors may have different effects in various domains. This observation suggests that the importance weights of different auxiliary feedback should not be determined manually or treated equally cross domains. **Second**, variable x_3 (i.e., remove) has small and negative effect on predicting the target feedback. This kind of negative feedback is unavailable in Sobazaar. The effect of negative feedback may be dependent on the reason to trigger such behaviors, which may vary from one user to another. It could be due to dislike, not interesting any more, replacement by a better alternative, or others.

To sum, different types of auxiliary feedback will have different impact on the target feedback cross domains. It is beneficial to involve them for better recommendations.

3.3. Data generation

A critical issue of top-N item recommendation is the lack of deterministic semantic interpretation for negative items in the dataset. In general, an unknown item may be due to dislike, or simply not recognized by a user. Fortunately, with the help of multi-type auxiliary feedback, we may lower the uncertainty to some extent and thus distill positive items from the unknown. To this end, we propose the following two ways.

(A) Linear regression: The factor analysis in Section 3.2 tells us in what situations a user is indeed likely to prefer an item, though she has no target action but only some auxiliary actions on the target item. For example, if a user has auxiliary actions x_1 and x_4 in Sobazaar, or x_1 and x_2 in Xing, it is probably that she may purchase the item eventually. Generally, the confidence that a user u may purchase an item j given some auxiliary feedback is determined by:

$$c_{u,j} = \beta_0 + \sum_{a=1}^{|A|} \beta_a x_a^{u,j}, \quad (2)$$

where $x_a^{u,j} \in \{0, 1\}$ is the value given by user u on item j in auxiliary dimension a . Based on Eqs. (1) and (2), we can predict if and to what extent a user will purchase an item, and thus treat this item as a positive item.

(B) Multi-dimension similarity: We assume that an item can be regarded as positive if it is closely similar in many dimensions to the other items purchased by the user. This assumption is much weaker in the previous recommendations since only one or two dimensions (auxiliary feedback) exist in the systems. With multi-type auxiliary feedback, item similarity can be computed more accurately. Specifically, for each kind of auxiliary feedback, traditional cosine similarity is employed to compute item similarity, given by:

$$\text{sim}(i, j)_a = \frac{\sum_u x_a^{u,i} \cdot x_a^{u,j}}{\sqrt{\sum_u (x_a^{u,i})^2} \sqrt{\sum_u (x_a^{u,j})^2}}, \quad (3)$$

where u is the user who has auxiliary feedback (in dimension a) on both items i and j . We also consider alternative similarity measures, such as Jaccard index⁵. Similar results are obtained without

³ https://en.wikipedia.org/wiki/Logistic_regression.

⁴ <http://www.cs.waikato.ac.nz/ml/weka/>.

⁵ https://en.wikipedia.org/wiki/Jaccard_index.

significant differences. Hence, the overall similarity cross dimensions is derived as follows:

$$\text{sim}(i, j) = \frac{\sum_{a=1}^{|A'|} \beta_a \cdot \text{sim}(i, j)_a}{\sum_{a=1}^{|A'|} \beta_a}, \quad (4)$$

where A' is the set of auxiliary feedback and target feedback, and the coefficient β_a for target feedback is fixed as 1. The ranking score is given as follows:

$$\text{rank}(j, u) = \sum_{i \in I_u} \text{sim}(i, j), \quad (5)$$

where I_u is the set of items purchased by user u . This ranking score is often adopted by item-based approaches for recommendations. In this case, the confidence that user u will purchase item j is given by:

$$c_{u,j} = \frac{\sum_{i \in I_u} \text{sim}(i, j)}{|I_u|}. \quad (6)$$

To avoid noise possibly derived from the data generation, we restrict to only adopt the data with confidence greater than a threshold, i.e., $c_{u,i} > \delta$, where $\delta \in (0, 1)$ is the threshold.

The confidence is an important factor to distinguish generated positive items from the original ones, whose confidence is the highest as 1. In next section, we will propose a confidence-aware ranking model for recommendation.

Discussion: Two points need to be highlighted. First, in our experiments, we use the two methods (A and B) sequentially. That is, we first find positive items by method A, and then search for more from the rest by method B. It is possible that an item can be identified if applying both methods simultaneously. In that case, the confidence from both aspects need to be merged together. We will leave this part as future work. Second, methods A and B are two possible ways to search for more positive items from the unknown ones, but not the only ones. Other alternatives could be devised. For example, we may base item correlations on their co-occurrence in different dimensions. Nevertheless, we'd like to stress that the essential is the use of correlations between auxiliary and target feedback learned from the factor analysis in Section 3.2.

4. BPR with generated data and confidence

In this section, we will elaborate the second stage of our approach, a variant of **Bayesian Personalized Ranking** model with **Generated data and confidence** (GcBPR) to accommodate both generated and original (positive) data as well as their confidence values.

4.1. Problem definition

Let $\mathcal{P} = [p_{u,j}]_{m \times n}$ represent a binary matrix of users' purchase over a number of items, where $p_{u,j} = 1$ indicates that user u purchased item j and 0 otherwise. The values m, n are the number of users and items, respectively. Let $\mathcal{A} = [x_a^{u,j}]_{m \times n \times |A|}$ represent a real-value tensor of users' interactions on items over auxiliary dimensions, where $x_a^{u,j}$ indicates the label user u has on item j in auxiliary dimension a . For simplicity, we reserve symbols u, v for users, and i, j, k for items. Hence, the task of item recommendation can be formulated as: given user-item purchase information $\mathcal{P}$ and associated auxiliary feedback tensor $\mathcal{A}$, generate a list of ordered items a user may have interested in.

In this work, we have generated more positive data from auxiliary feedback:

$$\tilde{p}_{u,j} = f(x_1^{u,j}, \dots, x_a^{u,j}, \dots, x_{|A|}^{u,j}), \quad (7)$$

where $f(\cdot)$ is a function to transform auxiliary feedback to target feedback. Two possible manners were introduced in Section 3.3 as

well as the confidence of generated data. Hence, the research problem can be reformulated as: given user-item purchase data $\mathcal{P} = [p_{u,j}] \cup [\tilde{p}_{u,j}]$ and the confidence of purchase data $\mathcal{C} = [c_{u,j}]$, generate a list of ordered items to active users.

4.2. The GcBPR model

The GcBPR model aims to optimize the ranking relationships between positive and negative items. Let θ be the model parameters, we intend to maximize the natural logarithm of posterior probability as follows:

$$\begin{aligned} \ln p(\theta | >_u) &= \ln p(>_u | \theta) p(\theta), \\ &= \ln \prod_{(u,i,j) \in \mathcal{D}} p(i >_u j | \theta) p(\theta), \end{aligned} \quad (8)$$

where $\mathcal{D} = \{(u, i, j)\}$ is a set of triplets, (u, i, j) means a user u purchased item i but not item j , and $p(i >_u j | \theta)$ is the probability that we restore the ranking relationships between items i and j for user u . In this work, the pair (u, i) may be either users' real actions or derived users' actions with possible confidence $c_{u,i}$. Hence, we compute the probability as:

$$p(i >_u j | \theta) = \sigma(\hat{p}_{u,i,j})^{c_{u,i}}, \quad (9)$$

where $\sigma(x) = 1/(1 + e^{-x})$ is a sigmoid function, and $\hat{p}_{u,i,j}$ is the ranking difference between pairs (u, i) and (u, j) . Note that we select the power function and take the confidence as the power value, to ensure that confidence has a sharper effect on the overall probability than other possible functions.

Hence, Eq. (8) can be formulated as:

$$\begin{aligned} \ln p(\theta | >_u) &= \ln \prod_{(u,i,j) \in \mathcal{D}} \sigma(\hat{p}_{u,i,j})^{c_{u,i}} p(\theta), \\ &= \sum_{(u,i,j) \in \mathcal{D}} c_{u,i} \ln \sigma(\hat{p}_{u,i,j}) + \ln p(\theta), \\ &= \sum_{(u,i,j) \in \mathcal{D}} c_{u,i} \ln \sigma(\hat{p}_{u,i,j}) - \lambda_\theta \|\theta\|^2, \end{aligned} \quad (10)$$

where λ_θ are regularization parameters to avoid over-fitting, $p(\theta)$ is subject to normal distribution with mean 0, and $\hat{p}_{u,i,j}$ is computed by:

$$\hat{p}_{u,i,j} = \hat{p}_{u,i} - \hat{p}_{u,j} = P_u^\top Q_i - P_u^\top Q_j, \quad (11)$$

where $P_u, Q_i, Q_j \in \mathbb{R}^d$ represent the latent feature vectors (in length $d \ll m, n$) of user u , item i , item j , respectively. The inner product of $P_u^\top Q_i$ is the estimated preference user u has on item i , i.e., $\hat{p}_{u,i}$. Thus, the model parameters θ to learn (by maximizing Eq. (10)) are the set of $\{P, Q\}$, where $P \in \mathbb{R}^{d \times m}$ and $Q \in \mathbb{R}^{d \times n}$ are the user-feature and item-feature matrices, respectively.

4.3. Model learning

The maximization of Eq. (10) is equivalent as the minimization of the negative of the same equation, which can be achieved by the Stochastic Gradient Decendent (SGD) approach. The learning procedure of our model GcBPR is given in Algorithm 1, which takes user-item (and generated) purchase and corresponding confidence matrices $\mathcal{P}, \mathcal{C}$ as input as well as parameters η, λ . First, all the variables are initialized by small random values (line 1). Second, we randomly draw a triple (u, i, j) from $\mathcal{D}$ as a training example, where i is an item purchased by user u , and j is not (line 3). The differences between estimated preference pairs (u, i) and (u, j) are computed by Eq. (11) (line 4). Third, the variables are updated according to the SGD rules (lines 5–7). The whole procedure is repeated until the objective loss is converged (line 2). Finally, all the variables are returned as output (line 8).

Algorithm 1: The GcBPR learning algorithm.**Input** : $\mathcal{P}, \mathcal{C}, \lambda, \eta$ (learning rate)

```

1 Initialize  $P, Q$  with random values in  $(0, 0.01)$ 
2 while not convergence do
3   randomly draw  $(u, i, j)$  from  $D$ 
4    $\hat{p}_{u,i,j} \leftarrow \hat{p}_{u,i} - \hat{p}_{u,j}$  by Equation (11)
5    $P_u \leftarrow P_u - \eta(c_{u,i}\sigma(-\hat{p}_{u,i,j})(Q_i - Q_j) + \lambda P_u)$ 
6    $Q_i \leftarrow Q_i - \eta(c_{u,i}\sigma(-\hat{p}_{u,i,j})P_u + \lambda Q_i)$ 
7    $Q_j \leftarrow Q_j + \eta(c_{u,i}\sigma(-\hat{p}_{u,i,j})P_u - \lambda Q_j)$ 
8 return  $P, Q$ 

```

The complexity of our GcBPR model is mainly taken by evaluating the objective loss. The time to compute the objective function (see Eq. (10)) is $O(d^*|D|)$, where $|D|$ is the size of training examples, and d is the number of latent factors. The values of $c_{u,i}$ can be computed offline in advance, and thus used as input to Algorithm 1. Hence, the complexity of our approach is linear with the size of training data. The running time⁶ of each iteration is 0.36 s, 2.1 s for Sobazaar and Xing respectively, which is comparable with the baseline methods.

5. Experiments

5.1. Experimental setup

All the data sets presented in Table 1 are used in our experiments. We split each of the datasets into two sub sets, where 80% are adopted as a training set and the rest 20% are used as a testing set, as described in Section 3. We adopt the following popular ranking metrics to evaluate the recommendation performance, including Precision (cut-off at 5, 10), Mean Reciprocal Rank (MRR) and normalized Discounted Cumulative Gain (NDCG). The precision measures the percentage of recommended items that are indeed preferred by users. The MRR considers the ranking position of the first hit (item recommended is indeed preferred). The NDCG takes further step to involve all the ranking positions of possible hits. In general, they evaluate different aspects of recommendation lists, and higher values indicate better performance. Due to space limitations, we refer readers to [15] for formal definitions of those metrics.

Competing methods: We have implemented and compared with the following approaches to justify the effectiveness of our approach. (1) BPR [16] is a well-known method for top-N item recommendation based on implicit feedback but involves no auxiliary feedback; (2) ABPR [10] considers additional one kind of auxiliary feedback; (3) TBPR [11] involves two kinds of auxiliary feedback; (4) MCBPR [12] makes use of all kinds of auxiliary feedback; (5) GcBPR is our approach with data generated from both linear regression and multi-dimension similarity as well as their confidence. The implementations of these approaches are based on (or provided by) the well-known recommendation toolkit LibRec.⁷

Parameter settings: The optimal settings of parameters are either determined by experiments or suggested by the literature. For simplicity, we fix the number of latent features as 20. For all competing methods, we apply grid search in $\{0.1, 0.01, 0.001, 0.0001\}$ to find the best learning rates (η) as well as regularization parameters (λ_θ). For ABPR: grid search for $\tau \in [100, 200]$ (step 100), $T \in [10, 20]$ and $T_0 \in [1, 5]$ (step 1) [10]; for TBPR: grid search for $\omega \in [0, 3000]$

Table 4

Ranking performance of all the GcBPR variants.

Dataset	Method	Pre@5	Pre@10	NDCG	MRR
Sobazaar	BPR	0.00885	0.00643	0.1149	0.0289
	GcBPR-l	0.0381	0.0227	0.2180	0.1433
	GcBPR-s	0.00898	0.00615	0.1191	0.0312
	GcBPR-c	0.0366	0.0220	0.2154	0.1394
	GcBPR	0.0390	0.0232	0.2231	0.1498
Xing	BPR	0.00128	0.00118	0.0461	0.00252
	GcBPR-l	0.0212	0.0120	0.1173	0.0784
	GcBPR-s	0.00456	0.00286	0.0573	0.0133
	GcBPR-c	0.0195	0.0115	0.1108	0.0710
	GcBPR	0.0215	0.0124	0.1179	0.0803

(step 200) and $\rho \in [0, 1]$ (step 0.2) [11]; for MCBPR: $\beta = 1$ to treat unobserved items as negative feedback on Sobazaar [12]. We refer readers to the previous papers for the definitions of those parameters.

5.2. Results and analysis

Comparison with GcBPR Variants. We have designed three variants to investigate each component of GcBPR, including (a) GcBPR-l: the GcBPR model with data generated from linear regression; (b) GcBPR-s: the GcBPR model with data generated from multi-dimension similarity; and (c) GcBPR-c: the GcBPR model with fixed confidence as 1. The results are presented in Table 4, where BPR is the baseline.

We can conclude that: (a) It is useful to generate positive data since both GcBPR-l and GcBPR-s reach better performance than BPR. (b) The data generated from auxiliary feedback (GcBPR-l) are more effective than those from similarity (GcBPR-s). (c) GcBPR outperforms GcBPR-l and GcBPR-s, indicating that more generated data lead to better performance. (d) GcBPR performs better than GcBPR-c, implying that data confidence is a useful factor to improve recommendations. In a summary, it is important to consider data generation and confidence in the GcBPR model, and the more data we can generate, the better performance we will reach. This phenomenon demonstrate that generating training data via a proper method is able to reduce the data sparsity problem.

Sensitivity analysis: In the method B, parameter δ determines if a generated positive record will be adopted as a training example or not. We adjust the values of δ in $(0, 1)$ with step 0.1, and illustrate the results in Fig. 1. Several observations can be noted: (a) small thresholds ($\delta < 0.2$) may bring in much noise and thus reduce the ranking accuracy; (b) proper values of δ (resp. 0.4 on Sobazaar, 0.5 on Xing) result in the best performance, though the relative improvements are small; (c) further increasing δ may significantly reduce the amount of positive data derived from similarity and mostly keep the data from linear regression (i.e., GcBPR-l). The effect of multi-dimension similarity will be gradually reduced (as on Sobazaar) or greatly wiped out (as on Xing).

Comparison with other methods: We compare the GcBPR model with a number of state-of-the-art approaches. The results are presented in Table 5 along with the percentage of relative improvements listed in the rows 'Improve'.

A number of conclusions can be drawn from the presented results. First, BPR consistently gets the poorest performance among all the competing methods across the two datasets. It means that adopting additional auxiliary feedback is beneficial to top-N item recommendations. Second, the performance gets better as the amount of auxiliary feedback increases: $BPR(0) < ABPR(1) < TBPR(2) < MCBPR(all)$ and $GcBPR(all)$, where the value in parentheses is the number of auxiliary feedback involved. It is clear that MCBPR and GcBPR outperform the others. Third, although both MCBPR and GcBPR consider all dimensions of auxiliary feedback,

⁶ Our experiments run on a PC with Intel Core processor having a speed of 2.2 GHz and 8GB of RAM.

⁷ <http://www.librec.net>.

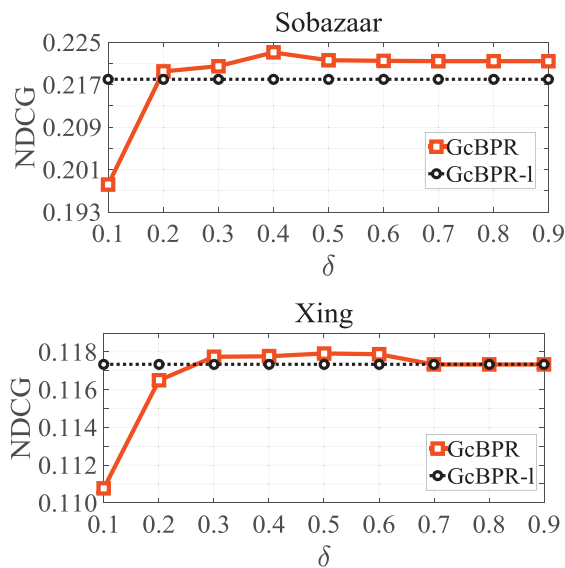


Fig. 1. The impact of parameter δ .

Table 5

Ranking performance of all the competing methods. The rows 'Improve' denote the percentage of improvements GcBPR has reached relative to MCBPR, i.e., between the best two approaches across the two datasets.

Dataset	Method	Pre@5	Pre@10	NDCG	MRR
Sobazaar	BPR	0.00885	0.00643	0.1149	0.0289
	ABPR	0.00885	0.00636	0.1159	0.0289
	TBPR	0.0153	0.0119	0.1574	0.0563
	MCBPR	0.0203	0.0140	0.1824	0.0791
	GcBPR	0.0390	0.0232	0.2231	0.1498
	Improve	92.12%	65.71%	22.31%	89.38%
Xing	BPR	0.00128	0.00118	0.0461	0.00252
	ABPR	0.00157	0.00133	0.0498	0.00303
	TBPR	0.0178	0.0121	0.1021	0.0620
	MCBPR	0.0192	0.0117	0.1071	0.0724
	GcBPR	0.0215	0.0124	0.1179	0.0803
	Improve	11.98%	5.98%	10.08%	10.91%

GcBPR consistently performs better than MCBPR. The average improvements GcBPR achieves relative to MCBPR are up to 67.38% on Sobazaar, and 9.74% on Xing. We attribute the improvements to the generation of additional positive data from multiple types of auxiliary feedback. Lastly, the improvements on Sobazaar are much greater than those on Xing. One possible explanation is that there are more dimensions of auxiliary feedback in Sobazaar (8) than that in Xing (3). More dimensions may result in better interpretation of target feedback and more accurate data generation.

To conclude, our GcBPR model obtains significant improvements over other counterparts across datasets, which verifies the fact that a better model manages to tackle data sparsity problem more efficiently.

6. Conclusion and future work

In this paper, we proposed a novel approach to exploit the value of multi-type auxiliary feedback in two stages. The first stage was to generate positive data from auxiliary feedback. Two generation methods were devised: one was predicted by linear regression, and the other by multi-dimension similarity. In addition, we de-

signed a confidence metric to evaluate the usefulness of generated data. In the second stage, we came up with a confidence-aware Bayesian personalized ranking model to accommodate both generated data and data confidence. Extensive experiments on two real-world datasets demonstrated the effectiveness of our approach in comparison with other well-performing methods.

For future work, we intend to study how data can be generated if auxiliary feedback is not linearly independent. In this case, non-linear methods are required to study the correlations between auxiliary and target feedback.

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